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12 MARCH 1956
SUPERSEDING
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MILITARY SPECIFICATION

WATCHES, WRIST (ALL GRADES)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers various types of wrist watches with durable and rugged characteristics, and accuracy as required for general purposes or precise missions, with either an off the center second hand or sweep second hand for military use throughout the various departments of the armed forces.

1.2 **Classification.** Military wrist watches have been graded according to accuracy and the following grades and types shall apply:

1.2.1 *Grades*

Grade I .....plus or minus 10 sec/day
Grade II .....plus or minus 30 sec/day
Grade III .....plus or minus 60 sec/day

1.2.2 *Types*

Type A — Watch with dial having a dull silver lustre finish and black markings, with closed minute track and off the center second hand (fig. 1).

Type B — Watch with dial having a dull silver lustre finish and black markings, with closed minute-second track and sweep second hand (fig. 2a).

Type C — Watch with dial having a black finish with white markings, with closed minute track and off the center second hand (fig. 1).

Type D — Watch with dial having a black finish with white markings, with closed minute-second track and sweep second hand (Fig. 2a).

Type E — Watch with dial having a black finish with white markings (24 hour system), with open minute-second track and sweep second hand (fig. 2b).

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

TT–R–58 — Radioactive Luminous Compound and Adhesives

FED SUP CLASS 6645

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MIL–S–3035 — Strap, Wrist Watch, Nylon

MIL–L–3918 — Lubricating Oil, Instrument, Jewel Bearing, Non-Spreading, Low Temperature

MIL–P–11446 — Packaging of Watches, Pocket, Stop and Wrist

MIL–F–13926 — Fire Control Materiel: General Specification Governing the Manufacture and Inspection of

MIL–P–17207 — Preservation and Packaging of Horological Repair Parts and Accessories Associated With Navigational Timepieces

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes

MIL–STD–129 — Marking for Shipment and Storage

MS35450 — Bearing, Jewel; Bar Hole

MS35451 — Bearing, Jewel; Center Bar Hole

MS35452 — Bearing, Jewel; Olive Hole

MS35453 — Bearing, Jewel; Olive Hole (cone cut)

MS35454 — Bearing, Jewle; Endstone

DRAWING

ORDNANCE CORPS

D7197100 — Case Assembly, Wrist Watch

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Qualification.** Watches furnished under this specification shall be of a product that has been tested and successfully passed the qualification tests specified herein (see 6.1.1 and 6.2).

3.2 **Design changes.** Subsequent to approval of a watch for a qualified product list, no changes shall be made in the design or construction of the movement or cases without approval from the contracting agency.

3.3 **Material.** All materials shall be of a uniform quality and free of any defects which might impair the functioning of the watch. Where a definite material is not specified it shall be in accordance with the best commercial practice and not affect the ability of the watch to meet the other requirements of this specification.

3.3.1 *Luminous material.* The luminous coating applied to the numerals and hands as specified herein shall be in accordance with Specification TT–R–58, Grade 65M, using Type A adhesive.

3.3.2 *Lubricant.* The lubricant used in the watch movement shall be in accordance with Specification MIL–L–3918.

3.4 **Movement**

3.4.1 *General.* The movement shall be of standard commercial manufacture and assembly, stem wind and stem set, of a size not

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less than 8/0 nor greater than 6/0 (sizes are in accordance with those established by English and American watch makers), and shall fit into the case snugly without interference. The winding stem shall be at the "3" o'clock position. All train wheels, pinions, plates and bridges of the movement shall be free of burrs, sharp edges, rough surfaces, and defects that would be detrimental to the functions of the unit. All steel parts and screws not protected by lubricant shall be black oxide finished, nickel plated or rustproofed in some approved manner except those parts the proper functioning of which would be detrimentally affected. If required, Grades I and II watches having a center pivoted sweep second hand, shall be provided with a suitable locking feature to stop the movement, permitting the watch to be set to the second (see 6.1.2).

3.4.2 *Mainspring.* The mainspring shall drive the completed movement a minimum of 36 hours without rewinding. Mainsprings shall be subjected to the mainspring cycle test and each shall meet an accelerated life test of 1200 cycles minimum without breakage and obtain a group average of not less than 1800 cycles. Failure to meet this requirement shall be cause for rejection of watches.

3.4.3 *Hairspring and balance wheel unit.* The movement shall have a temperature compensating hairspring and a nonmagnetic, solid, monometallic balance wheel. Both shall be made of materials such that the combination will not be permanently affected in the presence of a magnetic field.

3.4.4 *Regulator.* The regulator shall be at the center of adjustment within plus or minus 20 percent of its total range prior to and after completion of acceptance tests.

3.4.5 *Jewels.* Where jewel bearings are provided, bar hole, center bar hole, olive hole, olive hole (cone cut) and endstone jewel bearings shall be in accordance with the applicable military standards listed under 2.1.

3.4.6 *Setting and winding.* The crown and stem shall be assembled to provide waterproofing, and operation shall be smooth, without excessive torque. Continuous winding and setting operations shall not adversely affect the timekeeping and waterproof qualities of the watch.

3.5 Case assembly. The case assembly shall be fabricated in accordance with Drawing D7197100, and the component parts together with appropriate ring and crown shall be manufactured in accordance with applicable detail drawings.

3.5.1 *Crystal.* The first twenty (20) crystals made in accordance with the detail drawing of the case assembly drawing shall be forwarded to the Commanding General, Frankford Arsenal, Attention of Industrial Division, Fire Control Instrument Group, Philadelphia 37, Pennsylvania for approval prior to use in case assemblies. Crystals assembled to the completed watch shall be capable of meeting the storage temperature and waterproof tests prescribed herein.

3.5.2 *Strap.* The strap shall be manufactured in accordance with Specification MIL–S–3035 and shall be assembled to the completed watch prior to shipment.

3.6 Dial and hands

3.6.1 *Dial*

3.6.1.1 The dial shall be of nonferrous metal and in accordance with figures 1, 2a and 2b as applicable. The diameter shall be 0.994 within plus 0.000 or minus 0.003 inch with a minimum thickness of 0.015 inch. The dial, when secured to the movement, shall be such that the distance from the face of the dial to center line of the winding stem is 0.089 plus or minus 0.001 inch. The name of the manufacturer shall show on the face of the dial in an inconspicuous manner.

3.6.1.2 All markings shall be legible and shall have a baked, hard enamel paint coat-

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ing. The interior of the hour numerals shall be filled with the luminous material leaving a black or white outline on all types except Type E which shal have the hour numerals (1 to 12 only) painted solid with the luminous material, without the outline.

3.6.1.3 The minute track on all dials, except the dial Type E watches, shall be of a closed design, and each fifth minute graduation shall be marked heavier to designate the hour. The minute track on Type E dials shall be of the open design with each fifth minute graduation being heavier and painted solid with luminous material. On Types B, D and E watches, the minute track shall also serve as the second track for the sweep second hand and shall be appropriately marked as indicated in figures 2a and 2b.

3.6.1.4 The dial for Types A and C watches having an off the center second hand, shall have a circular recess, minimum 0.005 inch deep, to be as large in diameter as possible without interfering with the legibility of the "5" or "7", nor blocking out any part of the minute track at the "6" o'clock position. The second hand track shall be a closed design and follow the recessed portion of the dial shown in figure 1.

3.6.2 *Hands.*

3.6.2.1 The hour and minute hands shall have skeleton type features and be distinctly different in style and length. The hour hand shall be a heavy spade shape; the minute hand baton shape, and shall extend to at least the middle of the closed minute track but not beyond its outer circle. The interior portion of the hands shall be filled with the luminous material. (Hands are shown on fig. 3.)

3.6.2.2 The off the center second hand (fig. 4) shall have sufficient length to cover the closed second track but be within the recessed portion. It shall be of any style capable of providing for a dot of luminous material of such a size to be visible at normal reading distances. The place for the dot of luminous material shall be skeleton and shall be located at a point as close as possible to the end of the second hand without covering any part of the closed track.

3.6.2.3 The sweep second hand (shown in fig. 4) shall be a plain tapered counterbalanced type and shall have sufficient length to at least cover the minute track. The top surface of the sweep hand shall be painted with the luminous material for a sufficient length to be visible at normal reading distances.

3.6.2.4 The tips of the minute and sweep second hands shall be curved downward so that after assembly parallax is reduced to a minimum.

3.6.2.5 All hands shall be of steel; those used with watches Types A and B shall have a corrosion resistant black or blue finish, hands for watches Types C, D and E shall be nickel or rhodium plated.

3.6.2.6 In the finished watch, the hands shall be essentially parallel to the dial and to each other with sufficient clearance in each instance to prevent rubbing or binding. They shall rotate concentrically with respect to applicable minute-second dial track at assembly.

3.7 **Accuracy.** Watches shall meet the accuracy requirements in the dial up position and the crown down position. Daily rates shall be recorded for a period of three days in each position and the mean daily rate shall be determined therefrom. The mean daily rate shall not exceed the accuracy values for the respective grades and temperatures specified below:

	Temperature (Fahrenheit)		
	75°±3°	*40°±3°*	*125°±3°*
	Seconds	*Seconds*	*Seconds*
Grade I	±10	±30	±30
Grade II	±30	±60	±60
Grade III	±60	±120	±120

3.8 **Magnetism.** While running the watch shall not be adversely affected when sub-

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jected to a 10 gauss magnetic field for a period of at least five seconds and shall subsequently pass the final accuracy tests.

3.9 **Shock.** The watch shall show no evidence of damage affecting serviceability after a controlled drop from a height of one foot (equivalent to approximately a four foot free fall) at room temperature, once with the watch in the crown up position, and once with the watch in the dial up position as specified herein. The watch shall subsequently pass the final accuracy test.

3.10 **Vibration.** The watch shall be vibrated at an amplitude of 0.025 inch (total excursion 0.050 inch) and a frequency increasing from 0 to 30 vibrations per second and reversed to zero, in a cycle of approximately five minutes. The watch shall show no evidence of damage affecting serviceability as a result of vibration and shall subsequently pass the final accuracy tests.

3.11 **Storage.** The watch shall show no evidence of damage affecting serviceability when subjected to the ambient storage temperature of minus 80°F. and plus 160°F. in accordance with storage tests specified herein. The watch shall subsequently pass the final accuracy test.

3.12 **Waterproof.** When subjected to the waterproof test specified herein the completed watch shall show no evidence of leakage or other damage affecting serviceability and shall meet the accuracy requirements of the final accuracy test.

3.13 **Final accuracy.** Watches subjected to the environmental test of Section 4 shall meet the room temperature accuracy requirements specified in 3.7.

3.14 **Isochronism.** Watches shall be tested in the dial up position at room temperature in accordance with 4.4.17 except that the rate shall be recorded every six hours and the rate shall not vary more than two, three or five seconds for the Grade I, II or III watch respectively from the rate recorded in the previous six hour period.

3.15 **Identification.** Unless otherwise specified each watch shall show on the movement (barrel bridge or train bridge or both) the manufacturer's name and serial number and manufacturer's model or grade number. Each case shall be legibly marked on the exterior of the back with the applicable Ordnance grade number, i.e., "GRADE (I, II or III)" below the words "U. S. GOVERNMENT."

3.16 **Interchangeability.** Like parts shall be interchangeable in all like watches furnished by one manufacturer and shall not adversely affect timekeeping except for minor adjustments. Balance assemblies are considered interchangeable as an assembly.

3.17 **Workmanship.** Standards of workmanship shall be such that the watches shall meet all requirements herein and all components and subassemblies shall be in accordance with applicable workmanship requirements specified in MIL–F–13926.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Pilot lot.** A pilot lot representing initial production shall be submitted for complete test and acceptance in accordance with all requirements herein. The pilot lot size shall be determined by the Government inspector and the watches shall be representative of design, material and workmanship expected in normal production, applying best commercial practices.

4.2 **Sampling.**

4.2.1 *Qualification.* Ten sample watches and 10 sample mainsprings (with barrel and arbor) shall be submitted for approval as qualified products in accordance with 3.1.

4.2.2 *Production.* Sampling, including lot size, shall be in accordance with MIL–STD–105 and the acceptable quality levels specified herein except where special sampling is specified.

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4.3 **Inspection.** On watches submitted for qualified product approval or production acceptance, the Government inspector shall ascertain prior to and during assembly as well as after final assembly that production components, subassemblies and assemblies are in accordance with visual and dimensional inspection outlined below as well as the tests prescribed herein to verify conformance with all requirements of this specification.

4.3.1 *Visual and dimensional.*

4.3.1.1 *Material.* A visual inspection of component parts and assemblies shall be made to determine compliance with 3.3. Where defects or inferior quality is evident and the Government inspector deems a material analysis necessary, the contractor will be requested to submit samples or specimens to the contracting officer for analysis and approval. Where a technical analysis is desired, samples or specimens should be forwarded to the Commanding General, Frankford Arsenal, Attention of Industrial Division, Fire Control Instrument Group, Philadelphia 37, Pa.

4.3.1.1.1 *Luminous material.* Prior to production, certification of the lubricant shall be obtained in compliance with 3.3.1. Application methods and all markings, numerals and hands shall be inspected for neatness, legibility and adequacy in compliance with 3.6.1 and 3.6.2.

4.3.1.1.2 *Lubricant.* Prior to production, certification of the lubricant to be used in the watches shall be obtained in compliance with 3.3.2.

4.3.2 *Movement.* A visual and dimensional inspection shall be made to determine size and quality of workmanship (3.4.1) and design (3.2). Inspection shall be made on the pilot lot (4.1) and during phases of production as desired by the Government inspector.

4.3.2.1 *Jewels.* Where a movement contains jewels, one percent of the watches under contract, but not less than three, shall be examined to insure that the jewels are in compliance with 3.4.5 and the respective military standards.

4.3.3 *Case assembly.* Inspection for design and workmanship shall be made in accordance with 3.2, 3.5 and 3.17.

4.3.3.1 *Crystal.* The crystals shall be examined before and after assembly to the bezel and after the storage temperature tests (4.4.2.9) to determine compliance with 3.5.1.

4.3.3.2 *Strap.* Certification shall be obtained as to conformance with 3.5.2 and in addition a visual examination shall be made for workmanship.

4.3.4 *Dial.* The dial shall be inspected for size, markings, legibility, finish and type in accordance with the respective requirements of 3.6.1. Luminous coating shall also be inspected for adequacy of coverage and compliance with 3.3.1.

4.3.5 *Hands.* The hour, minute and second hands shall be inspected for style, length, shape and finish in compliance with 3.6.2. Luminous coating shall also be inspected for adequacy of coverage and compliance with 3.3.1.

4.3.6 *Regulator.* The regulator setting shall be checked after the final accuracy test (3.13) to determine compliance with 3.4.4.

4.3.7 *Identification.* All numbers and names shall be inspected for correctness, legibility and application in accordance with 3.15.

4.3.8 *Workmanship.* Quality of workmanship in conjunction with standard practices shall be inspected at the discretion of the Government inspector during in-process and on the end item to insure that watches are continually produced in accordance with 3.17.

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4.3.9 *Acceptable quality levels (AQL's) and classification of defects (CD's).* Unless otherwise specified, inspection shall be conducted in accordance with AQL's and CD's

TABLE I. *Classification of Defects*

Major	*AQL 2.5*	*Requirement*	*Test procedure*	*Method of inspection*
1.	Setting	3.4.6	4.4.2.3	Functional
2.	Winding	3.4.6	4.4.2.4	Functional
3.	Dial	3.6.1	4.3.4	Visual
4.	Hand	3.6.2	4.3.5	Visual and Functional
5.	Accuracy (75°F.)	3.7	4.4.2.5	Functional
Major	*AQL 4.0 Inspection level L–8 of Table VII*	*Requirement*	*Test procedure*	*Method of inspection*
1.	Accuracy (40°F. and 125°F.)	3.7	4.4.2.5	Functional
2.	Shock	3.9	4.4.2.7	Functional
3.	Vibration	3.10	4.4.2.8	Functional
4.	Storage	3.11	4.4.2.9	Physical
5.	Crystal	3.5.1	4.3.3.1	Visual
6.	Waterproof	3.12	4.4.2.10	Visual
7.	Final accuracy	3.13	4.4.2.11	Functional
Minor	*AQL 4.0*	*Requirement*	*Test procedure*	*Method of inspection*
1.	Case assembly	3.5	4.3.3	Visual
2.	Strap	3.5.2	4.3.3.2	Visual
3.	Regulator	3.4.4	4.3.6	Visual
4.	Identification	3.15	4.3.7	Visual
5.	Workmanship	3.17	4.3.8	Visual

specified in table I for all visual and dimensional inspection and functional tests. AQL's specified in table I are expressed in percent defective.

4.4 Test procedures

4.4.1 *Qualification tests*

4.4.1.1 *Mainspring tests.* Mainsprings with barrels and arbors shall be tested in accordance with 4.2.1. and 4.4.2.1.

4.4.1.2. *Completed watches.* Qualification tests (4.2.1) shall consist of all tests specified herein and after completion of all tests the watches shall be run for a period of one (1) year (6 months with dial up and six months with crown down). At the conclusion of this period the watches shall be tested in accordance with room temperature requirements of 3.7.

4.4.1.3 *Regulation.* At the end of the 1 year running period, the watches may be regulated before performing room temperature accuracy tests in compliance with 4.4.1.2. The contractor may be permitted to have a representative perform regulation under supervision of the Government inspector or grant permission to allow the Government inspector to regulate the watches at the laboratory. In either event, only regulation shall be permitted to improve the daily rate for acceptance after 1 year running.

4.4.2 *Acceptance tests.* Acceptance tests shall be made on individual lots submitted for acceptance in accordance with 4.2.2 and 4.3.8.

4.4.2.1 *Mainspring test.* One half of one percent but not less than four mainsprings produced from each run of material from each heat treatment lot used in production watches under procurement shall be subjected to the cyclic tests to determine compliance with 3.4.2. Each spring shall be assembled in barrel with arbor. Half the springs shall be stored under room conditions (75°F. plus or minus 5°F.) for a period of 14 days, the other half of the springs shall

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be stored at 120°F. (plus or minus 3°F.), 100 percent relative humidity for a period of 14 days. After this storage period, each spring shall be fully wound and then permitted to unwind to simulate a 24 hour wind. The entire wind and unwind cycle shall be accomplished in approximately 60 seconds.

4.4.2.2 *Hairspring and balance wheel unit.* The hairspring and balance wheel unit shall be considered acceptable for compliance with 3.4.3 if the watch is capable of meeting the requirements of 3.7, 3.8 and 3.14.

4.4.2.3 *Setting tests*

4.4.2.3.1 *Hour-minute setting.* Six settings shall be made in two hour increments at each testing temperature to insure compliance with 3.4.6 and 3.6.2.6.

4.4.2.3.2 *Second setting.* Where a watch is required to have a second setting mechanism (6.1.2), at least five different settings at each testing temperature shall be made to insure that a precise stop and start action can be obtained without adverse effect on the hands or movement in accordance with 3.4.1 and 3.4.6.

4.4.2.4 *Winding test.* The accuracy test specified below shall be utilized during each daily winding to determine compliance with 3.4.6 and excessive torque shall be interpreted as any additional torque that is necessary to wind watches as compared to the normal winding practice used in the trade.

4.4.2.5 *Accuracy test.* Prior to the accuracy tests, watches shall be conditioned by being fully wound and shall have run a minimum of 36 hours without rewinding to determine compliance with 3.4.2. During the conditioning period the watches shall be subjected to the test temperature for at least four hours prior to the test. The watches shall be wound at the beginning of each test and each 24 hours thereafter for the duration of the tests. The watches shall be recorded every 24 hours and shall be rejected if the mean daily rate exceeds the requirements of 3.7.

4.4.2.5.1 Temperature tests at plus 40°F. and plus 125°F. shall be conducted under controlled ambient conditions and the watches may be gradually exposed to the specified temperatures prior to actual testing and subsequent thereto.

4.4.2.6 *Magnetism test.* One percent of the watches under contract but not less than three watches shall be placed in a magnetic field generated by conventional laboratory methods in compliance with 3.8.

4.4.2.7 *Shock test.* For the crown up position, the watch without strap shall be positioned and secured to a steel block in such a manner that the area of the rim of the case between "8" and "10" o'clock shall be in full contact with the locating surface of the steel block with pressure applied to a portion of the rim on each side of the pendant to prevent shifting at the moment of impact. For the dial up position, the watch without strap shall be secured in such a manner that the case cover shall rest on the locating surface of the steel block, and a ring of pressure shall be applied on the bezel to prevent shifting at the moment of impact. When the watch is properly positioned and secured, the steel block shall be dropped onto a horizontal surface of dry true hickory, grain up, which is supported by a two inch steel plate. The weight of the steel mounting block shall be 1.75 pounds plus or minus one ounce and of the area of contact at the instant of impact with the hickory shall be 1.25 minus 0.1 square inch. The dimension of the hickory shall be 2¼ inches in diameter by 3½ inches thick. During the operation of the test care should be exercised to properly guide the block (with watch) to prevent any gyration.

4.4.2.8 *Vibration test.* The watch shall be vibrated in accordance with 3.10 as follows:

30 minutes with vibration perpendicular to dial.

30 minutes with vibration in plane of dial and in direction from 12 to 6.

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30 minutes with vibration in plane of dial and in direction from 9 to 3.

4.4.2.9 *Storage.* In compliance with 3.11, subject the watches to ambient temperature as follows:

Store at minus 80°F. (plus or minus 2°F.) for 24 hours.

Store at room temperature (75°F. plus or minus 3°F. for 24 hours.

Store at plus 160°F. (plus or minus 2°F.) with at least 50 percent relative humidity for 24 hours.

Store at room temperature for 24 hours.

Note. Watches shall not be run during storage tests. Temperature changes in the watch may be gradual to avoid thermal shock.

4.4.2.10 *Waterproofing test.* The watches shall be subjected to one of the following test methods to determine compliance with 3.12. With the exception of final accuracy test (3.13) the waterproofing test shall be the last test to which the watches shall be subjected before delivery.

4.4.2.10.1 *Method I.* The watch shall be suspended in a vessel of water containing approximately one percent of Aerosol by weight and placed in a chamber which shall be evacuated to a negative pressure of 5½ inches of mercury for a period of 15 seconds. Watches showing evidence of air leakage during the test period shall be rejected.

4.4.2.10.2 *Method II.* The watch shall be suspended in a pressure vessel above water containing approximately one percent of Areosol by weight and subjected to an air pressure of 5½ inches of mercury for a period of 45 seconds minimum. The watch shall then be submersed in the water with the subsequent release of air pressure. Watches showing any evidence of air leakage for a period of 15 seconds or less after release of the air pressure shall be rejected.

Note. In either method leakage will appear as bubbles emanating from the crystal or stem of the watch. Bubbles which are the result of entrapped air on the various exterior parts of the case shall not be considered as a leak. The watches being tested and the water used shall be at room temperature. Distilled water shall be used.

4.4.2.11 *Final accuracy test.* On completion of tests in 4.4.2.5 to 4.4.2.10 inclusive, the watches shall meet the room temperature accuracy requirements of 3.7.

4.4.2.12 *Isochronism test.* This test shall be in accordance with 4.4.2.5 except that the rate shall be recorded in compliance with 3.14.

4.4.2.13 *Interchangeability tests.* The contracting officer should schedule interchangeability tests when it is deemed necessary to insure compliance with 3.16.

4.4.2.14 *Resubmission after rejection.* Watches rejected for failure during or after testing may after readjustment and correction of defects, be resubmitted for complete test. Watches rejected on a retest shall not be resubmitted without specific approval of the contracting agency.

5. PREPARATION FOR DELIVERY

5.1 **Packaging and packing.** Military, limited military and minimum military, preservation, packaging and packing shall be in accordance with the requirements of Specification MIL–P–11446 and applicable protection levels (A, B or C).

5.1.1 *Repair parts and accessories.* Repair parts and accessories shall be preserved and packaged in accordance with MIL–P–17207.

5.2 **Marking.** Marking shall be in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Ordering data.** Procurement documents shall specify the following:

(a) Title, number and date of this specification.

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(b) Grade and type required (see 1.2.1 and 1.2.2).

(c) Packaging level (A, B or C), Packing level (A, B or C).

(d) Repair parts or accessories (5.1.1).

(e) Special identification (3.15).

6.1.1 Until such time as a Qualified Product List is established for jeweled or non-jeweled watches in accordance with 4.4.1.2, the following minimum number of jewels with respect to the grade indicated should be specified in the contract:

Grade I —17 Jewels
Grade II —15 Jewels
Grade III— 7 Jewels

6.1.2 In compliance with 3.4.1, the contract should specify if the locking feature is desired in the Grade I or Grade II watch to stop the movement for second setting purposes.

6.2 Qualification remarks. In the procurement of products requiring qualification the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Army-Navy-Air Force Qualified Products List. The attention of suppliers is called to this requirement, and manufacturer's are urged to communicate with the Commanding General, Frankford Arsenal, Attention of Industrial Division, Fire Control Instrument Group, Philadelphia 37, Pa., and arrange to have the wrist watches that they proposed to offer to the Army, Navy or Air Force tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification.

6.3 Dial type "E" is normally for U. S. Air Force use.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Ordnance Corps
Navy—Bureau of Ships
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Other interests:
Army—TE
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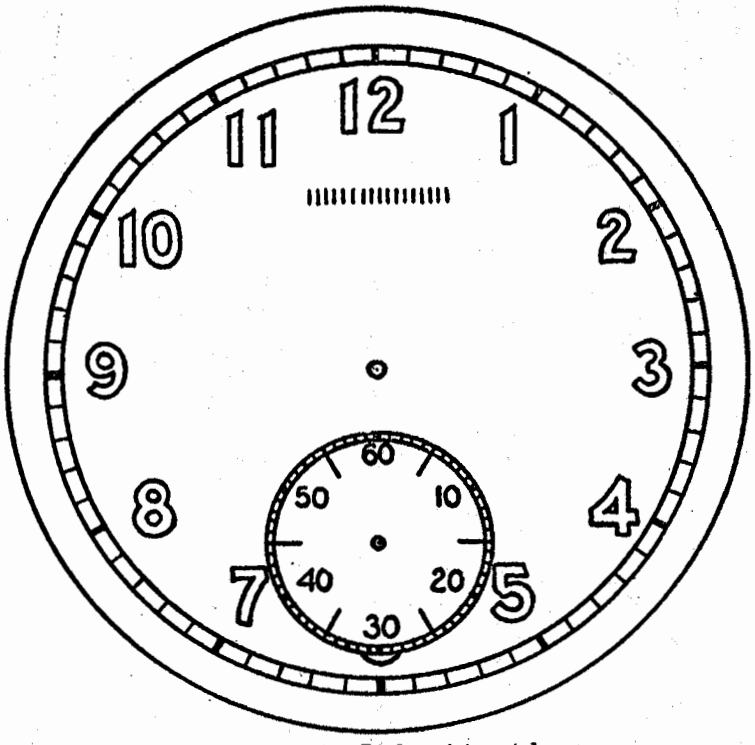


FIGURE 1.—*Dial, wrist watch.*

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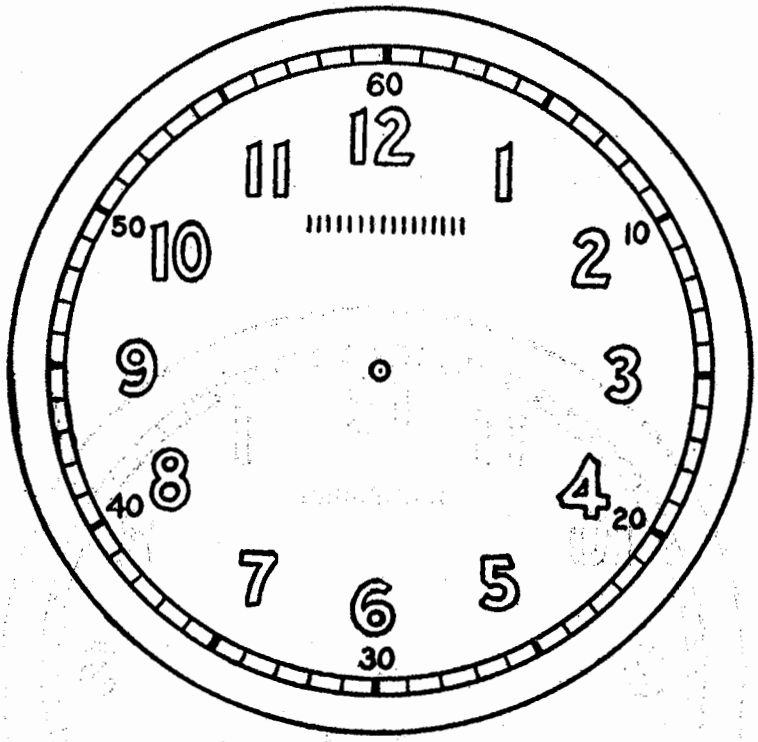


FIGURE 2.—*Dial, wrist watch.*
a

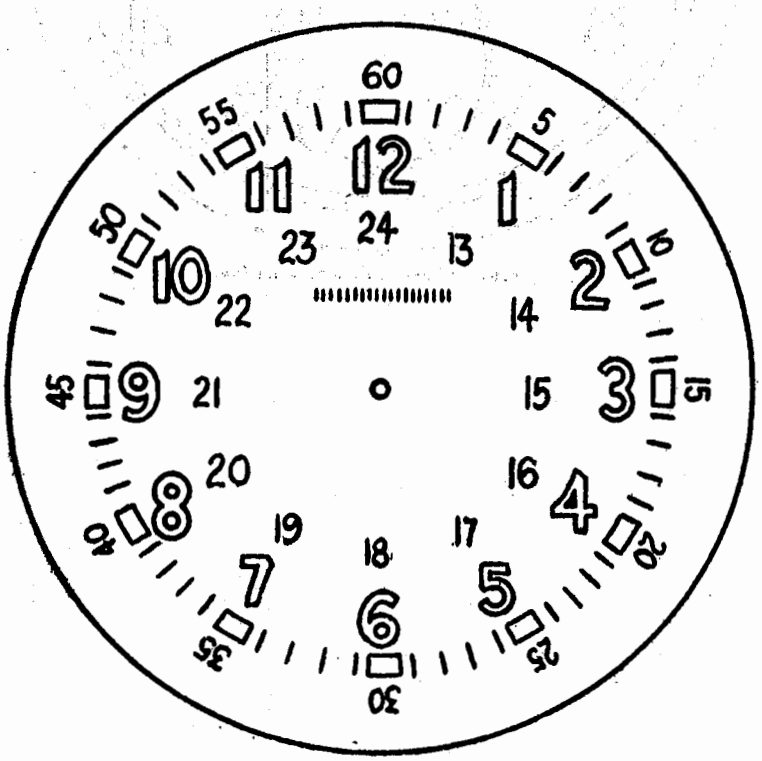


FIGURE 2.—*Dial, wrist watch.*
b (24-hour)

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BATON SHAPE



HEAVY SPADE SHAPE

FIGURE 3.—*Hands, wrist watch.*

SWEEP SECOND HAND



OFF THE CENTER SECOND HAND



FIGURE 4.—*Hands, wrist watch*

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